

ITSS SOFTWARE DEVELOPMENT/ SOFTWARE DESIGN AND CONSTRUCTION

Lab 08: User Interface Design

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1. SUBMISSION GUIDELINE

To submit this lab, you and your teammate have to push all the work to the group GitHub repository, following the naming convention `MSTeamsName-TeamNumber` (e.g., TKXDPM.VP.20251-01).

For this lab, you are required to checkout a branch named `release/lab08`, which contains all tasks related to user interface design.

The deadline for each submission will be updated by the lecturer or the teaching assistant. Therefore, you need to pay attention to avoid late submissions.

2. USER INTERFACE DESIGN

When the actor is human, boundary classes can be utilized to capture the requirements on a user interface. The interactions between people and systems might be through various kinds of user interfaces such as batch interface (BI), command-line interface (CLI), and graphical user interface (GUI) (with GUI, depending on the type of product you are developing, e.g., a web application, a desktop application, or a mobile application..., there may also be specific characteristics unique to each.).

In this section, we use GUI to illustrate how to design UIs step by step. You can have a look at some samples of UI/UX design (which was thoroughly studied in the *User Interface and User Experience* course):

- Charity Donation App: <https://duchai.framer.website/Chado>
- Re-design Starbuck Ordering App: <https://duchai.framer.website/Redesign-Starbuck>

All the images/diagrams in this lab are merely examples, so *they can be inaccurate to the problem statement*. You need to check carefully to ensure that your work are consistent with your use case model.

2.1. STANDARDIZING SCREEN DESIGN

The design and standardization of screens play a crucial role in ensuring usability and consistency across different products. Depending on the type of application, the focus may vary. For instance, a web application often emphasizes cross-platform compatibility and responsive layouts, while a desktop application may prioritize performance and efficient use of system resources. Regardless of the platform, applying standardized screen design principle is essential to create a cohesive user experience and minimize issues during both the design and development stages.

Screen Layout

Screen layout defines the arrangement of interface components such as titles, menus, content areas, and action buttons. Standardizing the screen layout ensures design consistency and provides users with an intuitive and easy-to-use experience. Screen layout serves as the foundation for creating wireframes, as it defines the placement and roles of key components on the screen. With a clear screen layout, wireframes can be designed more consistently and effectively.

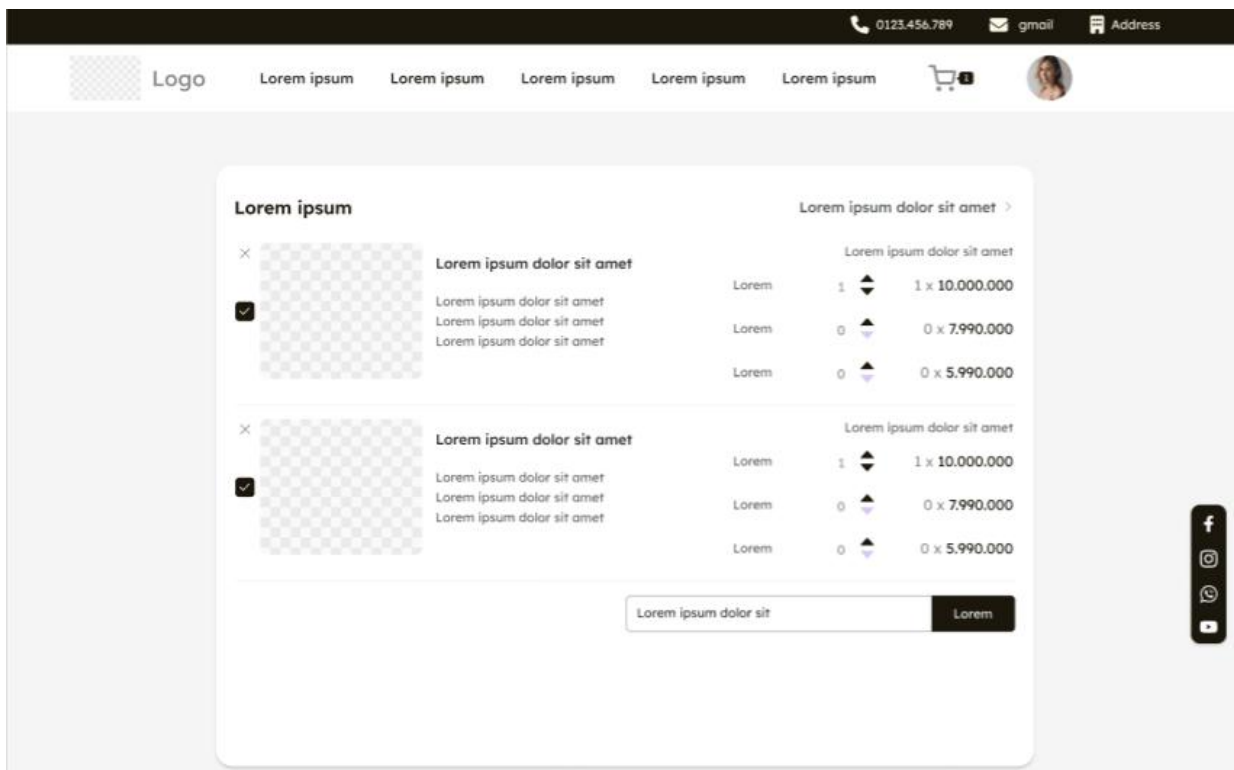


Figure 1. A sample for wireframe (cart screen) in web application

- **Minimum resolution:** 1366 × 768 pixels, for example, with adaptability for higher resolutions or different aspect ratios (e.g., mobile or tablet). Ensure layouts adapt gracefully to different screen sizes (responsive design).
- **Location of standard buttons:** Positioned in a consistent and recognizable location, typically at the bottom (vertically) and aligned center or right horizontally.
- **Location of screen title:** Titles should be placed at the top center of the screen for consistency.
- **Location of menu bar:** It is recommended to place the menu bar at the top-right corner of the screen for easy access and consistency.

UI Kit

UI Kit is a collection of reusable interface elements, such as buttons, icons, input fields, and navigation components, combined with visual styles like color palettes and typography. It provides designers with ready-to-use building blocks that ensure consistency and speed up the design process across different screens and projects.

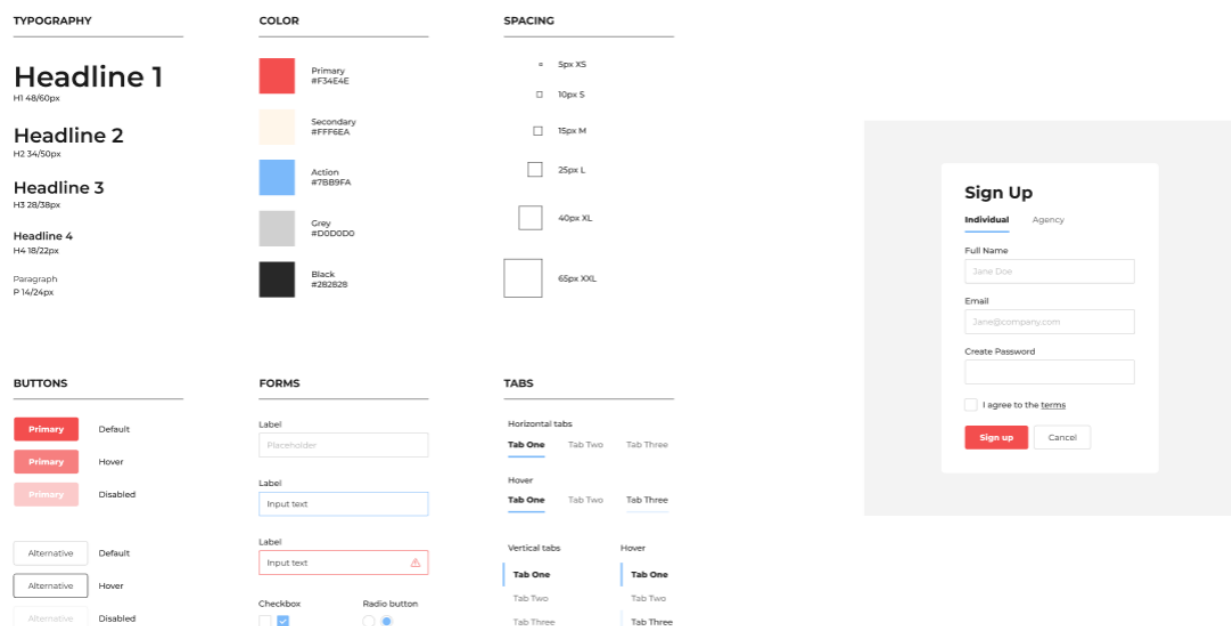


Figure 2. A sample for UI Kit (Source: Figma)

- **Font family:** Choose a single font family to be used consistently across the entire application. The font should be widely available and legible (e.g., Segoe UI, Roboto, or an equivalent).
- **Font sizes:** Define a clear hierarchy of text sizes, for example: main title (Heading1, e.g., 24px), sub-title (Heading 2, e.g., 20px)...

- **Color palette:** Establish a standardized set of colors to be used throughout the application, including primary color (e.g., #0078D7), accent/highlight color (e.g., #FF5722), neutral/background colors for text and layout balance...
- **Icons & graphics:** Use a consistent icon set (e.g., Material Icons, FontAwesome). Ensure images maintain correct aspect ratio and quality.

Data Formatting

Data formatting defines how information is structured and presented within the user interface. Standardizing data formats, such as dates, numbers, currencies, and input fields, ensures consistency, improves readability, and reduces user errors.

- **Numeric values:** Use comma separators every three digits (e.g., 1,000,000).
- **Currency format:** Always show currency symbols or units (e.g., \$1,000, 1,000đ, 1,000 VND).
- **Date & time format:** Use a standardized pattern (e.g., DD/MM/YYYY, YYYY-MM-DD, or DD-MM-YYYY...).
- **String handling:** Display only valid characters; prevent rendering of unsupported or unsafe characters.

Control

Control design defines how users interact with the system through inputs, navigation, and screen transitions. Establishing clear rules for input validation, focus navigation, and keyboard interactions ensures both usability and consistency across the application.

- **Input validation process:** First, check whether the input field is empty. Then, verify whether the input format is valid.
- **Focus navigation sequence:**
 - Input fields should follow a logical sequence, supporting navigation via keyboard/tab where applicable.
 - Secondary screens (e.g., help or manual screens) should appear as popups, preventing interaction with the main screen until they are closed.
 - The application typically starts with an opening screen, then a splash screen, followed by the home screen.
- **Keyboard and navigation input:**
 - No shortcuts are required.
 - Provide back buttons for returning to the previous screen.

- A close button (“X”), placed at the familiar position (e.g., top-right corner for desktop), should be available to exit a screen.

Other standardization

- Error handling: Clear and concise error messages should be displayed to inform users of the problem and, if possible, guide them on how to resolve it.
- ... (you can define your own standardization here)

The items above are merely examples for standardizing screen design. You should discuss them thoroughly with your teammates and then define concrete, project-specific standards. These standards will be submitted in a PDF file named **ScreenStandardizationRequirements**.

2.2. CREATING MOCKUPS

Creating mockup designs can be done using various applications such as Figma (highly recommended), InVision Studio, Paint, Adobe graphic design software, AdobeXD, or Scene Builder. You may use any type of tool to create mockup designs. Afterwards, export the screens as corresponding image files (in PNG or JPG format) and make sure to name them clearly for easy identification.

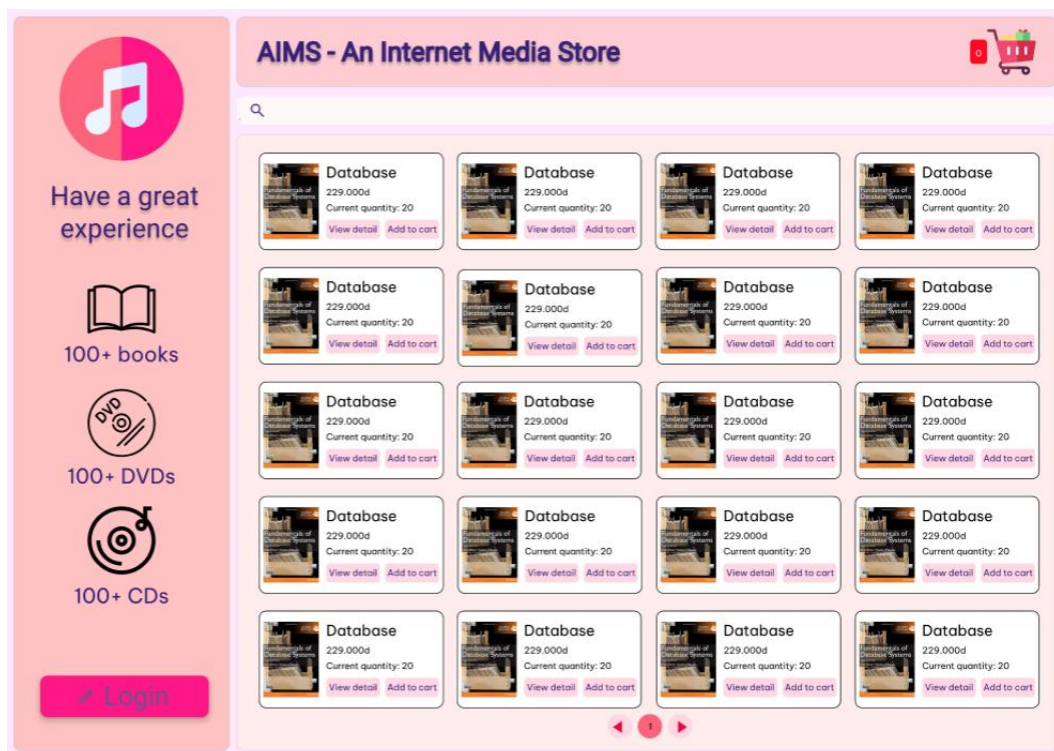


Figure 3. A sample mockup for AIMS homepage (for desktop application)

Figure 4. A sample mockup for AIMS cart screen (for desktop application)

Figure 5. A sample mockup for AIMS delivery information screen (for desktop app)

2.3. SCREEN TRANSITION DIAGRAM

In this part, you are required to create screen transition diagrams using draw.io or any drawing tool (including Figma) as specified in the previous section, for the following use cases: browse/search/filter/sort products, view product details, view/manage cart, and place order (including payment). Afterward, combine all individual diagrams into a single consolidated diagram that illustrates the complete flow of transitions between screens, making it easier to understand relationships and dependencies across the system. Finally, export the result as an image file (PNG or JPG format) named **ScreenTransitionDiagram** for submission.

For example, on the homepage, when a user clicks on any product, the system displays its detail page in a popup. At this point, the homepage becomes non-interactive until the popup is closed. Therefore, the relationship between the homepage and the detail page is defined as a “transition to a dependent child screen”.

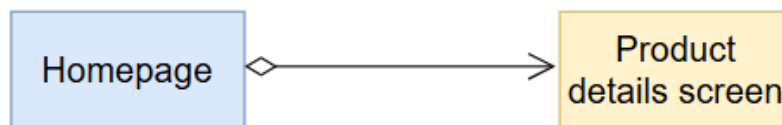


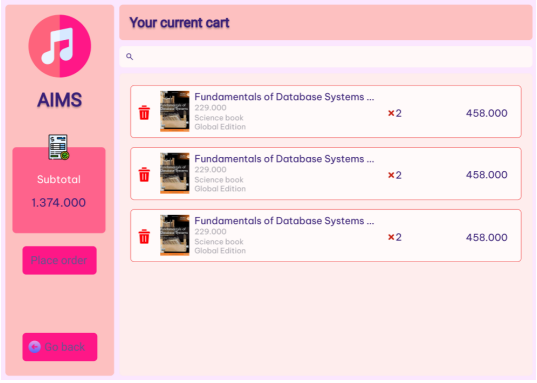
Figure 6. A sample for screen transition diagram

2.4. SCREEN SPECIFICATION

For each screen you have designed above, you are required to prepare a screen specification table according to the template below. Discuss thoroughly with your team to divide tasks appropriately. After completing the specifications, submit all documents combined in a single PDF file named **ScreenSpecifications**.

Screen specification

AIMS Software	Date of creation	Approved by	Reviewed by	Person in charge
Screen specification: Cart Screen	20/3/2025	DungVT	NgocNT	AnhTV
	Control	Operation	Function	
	Area for displaying the subtotal	Initial	Display the subtotal	

	Area for display items in the cart	Initial	Display the media with the corresponding information
	Place order button	Click	Display the Delivery form
	Delete button	Click	Remove the item from the cart

Defining the field attributes

Screen name: View cart				
Item	Length	Type	Color	Remarks
Media title	50 characters	Text	Blue	Left-justified
Price	20 digits	Numeral	Gray	Left-justified
Subtotal	20 digits	Numeral	Gray	Left-justified

2.5. SUBMISSION REQUIREMENTS

Now, you will complete the tasks (from section 2.1 to section 2.4) step by step as instructed. Each member will be responsible for certain screens, according to the group's assignment. To submit your work, you need to create a directory named **DetailedDesign/UserInterfaceDesign**, then place all the tasks you have completed into it. You may refer to the following structure:

```
DetailedDesign/
├── UserInterface/
│   ├── ScreenStandardization.pdf
│   ├── ScreenTransitionDiagram.png
│   ├── ScreenSpecifications.pdf
│   └── Mockups/
│       ├── Homepage.png
│       └── ... (put your screen images here)
```